An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

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Outline

1. The question
2. Main results
3. Natural approaches that fail
4. Technical overview
5. Further results
6. Conclusion

FO^3 is a singular point

Three-variable first-order logic restricted to binary relations (FO^3) is already a singular point among the finite-variable fragments FO^k.

	FO^2	FO^3
triangle, transitivity		expresses them
over words		expresses all of first-order logic
(finite) satisfiability	decidable; finite models suffice	undecidable; **can force infinite models**
property testing		a single $\forall\exists\forall$ sentence is not testable
symmetric model counting	polynomial for every sentence	$\#\mathrm{P}_1$-complete for a fixed sentence
spectra		Asser's question reduces to spectra of FO^3 sentences

satisfiability: even over a single binary relation; property testing: with a constant number of queries; Asser's question (are first-order spectra closed under complement?) is equivalent to $\mathrm{NE} = \mathrm{coNE}$

Three variables even suffice for all of mathematics: Tarski formalized set theory in $\mathcal{L}^\times$, a variable-free logic of equations between terms of the calculus of relations (CoR), and these equations have exactly the expressive power of three-variable sentences (Tarski and Givant 1987).

Model checking: FO^3 is singular here as well

Model checking is solvable in polynomial time for every FO^k: a k-variable formula φ can be evaluated naively in time $O(|\varphi| n^k)$ (n elements).

$\mathrm{FO}^k,\ k \geq 4$	not improvable to $O(n^{k-\varepsilon})$ under the hyperclique hypothesis, even for a fixed sentence over a single binary relation (Nakamura and Uezato 2026, Corollary 32)
FO^3	every sentence in time $2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ $\omega < 2.3714$ (Nakamura and Uezato 2026, Theorem 10)

FO^3 and CoR define the same binary queries (Tarski and Givant 1987), and the key is the central operation of CoR, **composition**:

- it eliminates the third variable, $\exists z\,(P(x,z) \wedge Q(z,y)) \iff (P\,;Q)(x,y)$;
- it is a Boolean matrix product, computable in subcubic time $O(n^\omega)$.

The factor $2^{O(|\varphi|)}$ comes from the translation into CoR.

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Is the blowup necessary?

- Whether it is necessary sounds like a modern question about query compilation, but it is as old as Tarski's translation from the 1940s (Tarski and Givant 1987), and its roots reach back to the nineteenth-century calculus of relations.
- It remained open (Fletcher et al. 2015, Remark 4.5; Nakamura 2022, §8; Nakamura and Uezato 2026, Question 3), although under the 3-OV hypothesis no translation can be computed in polynomial time (Nakamura and Uezato 2026, Corollary 40).

We settle it: **Tarski's translation is optimal in the worst case.**

Writing T_φ for a smallest CoR term equivalent to φ on finite structures,

$\varphi \Longrightarrow T_\varphi \qquad |T_\varphi| = 2^{O(|\varphi|)}$ (classical), $\qquad |T_\varphi| = 2^{\Omega(|\varphi|)}$ (our result).

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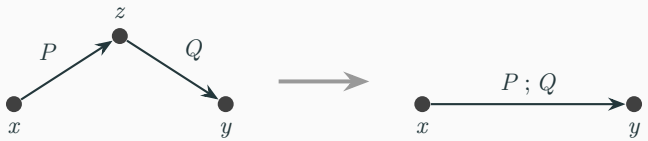
The calculus of relations (CoR)

On a structure with nonempty domain $\mathbb{D}$, CoR uses the constants

$$\mathbf{0} \coloneqq \emptyset, \qquad \mathbf{1} \coloneqq \mathbb{D}^2, \qquad \mathbf{id} \coloneqq \{(u,u) : u \in \mathbb{D}\},$$

together with union, intersection, complement $\overline{X} \coloneqq \mathbb{D}^2 \setminus X$, converse $X^\smile \coloneqq \{(v,u) : (u,v) \in X\}$, and composition

$$X \,;\, Y \coloneqq \{(u,v) : \text{some } w \in \mathbb{D} \text{ satisfies } (u,w) \in X,\ (w,v) \in Y\}.$$



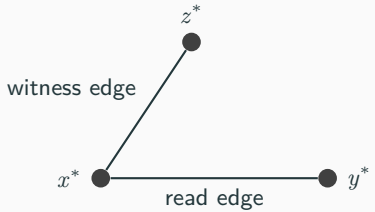
Composition implements the elimination step $\exists z\,(P(x,z) \wedge Q(z,y)) \iff (P\,;\,Q)(x,y)$; a vertex z with $P(x,z)$ and $Q(z,y)$ is a **witness** for (x,y).

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The formula G_k

$$G_k(x, y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[(P_i(x, y) \wedge Q_i(x, z)) \vee (Q_i(x, y) \wedge P_i(x, z)) \right] \qquad \text{one quantifier, size } 8k$$



At every **coordinate** i, one of the two edges **meets** P_i and the other meets Q_i, in either order. Then the read edge **accepts** the witness edge.

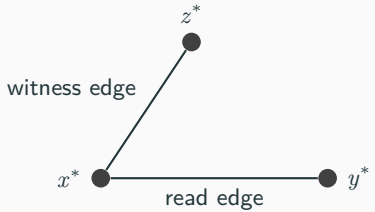
DNF: fix the order at every coordinate. For $A \subseteq [k]$, $H_A \coloneqq \bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i$; a read edge in H_A accepts every witness edge in H_{A^c}:

$$G_k \equiv \bigcup_{A \subseteq [k]} \left[H_A \cap (H_{A^c} ; \mathbf{1}) \right] \qquad 2^k \text{ compositions}$$

row projection: $X ; \mathbf{1}$ holds at (x, y) iff x has an outgoing edge in X

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Main theorem

A CoR **term** T is a syntax tree; $\#_{;}(T)$: its compositions.
A **relational circuit** C (DAG) shares subterms; $g_{;}(C)$: its distinct composition gates.
$\equiv_{\mathrm{fin}}$: equivalence on all finite structures over the input signature.

Thm. (Exponential lower bounds for full CoR)

For every $k \geq 1$, every relational circuit $C \equiv_{\mathrm{fin}} G_k$ has $g_{;}(C) \geq 2^{k-1}$, and every CoR term $T \equiv_{\mathrm{fin}} G_k$ has $\#_{;}(T) \geq 2^{k-1}$.

- Since G_k has $8k$ nodes: $2^{\Omega(k)}$ compositions from a source of size $O(k)$, **even when all other operations are free and arbitrarily numerous**.
- Single exponential: sufficient and necessary for FO^3-to-CoR compilation, even with sharing.
- Counts **whole-matrix products** rather than scalar operations (on our structures, G_k is evaluable in linear time).

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Term size, and where the blowup begins

Thm. (Term size)

Every CoR *term* $T \equiv_{\text{fin}} G_k$ *has at least* $2k2^k - 1$ *nodes, and* G_k *is equivalent to a* CoR *term with* $(2k+6)2^k - 5$ *nodes.*

A finer count: singling out one of the 2^k sets A examines all k coordinates.

$c_n(G_k)$: least $g_{\cdot}(C)$ of a circuit computing G_k on all n-element structures;
$n_0(N) \coloneqq \lceil 12N^2 \ln(2N) \rceil + 3$.

Prop. (Domain size)

$$\max\{2^{j-1} : 1 \le j \le k,\ n \ge n_0(2^j)\} \ \le\ c_n(G_k) \ \le\ \min\{2^k,\ 4kn\}.$$

In particular, $c_n(G_k) \ge 2^{k-1}$ *for* $n \ge n_0(2^k) = O(k4^k)$, *and* $c_n(G_k) < 2^{k-1}$ *for* $n < 2^k/(8k)$.

The blowup begins at domain size $2^{\Theta(k)}$. All these bounds follow from one lemma about a single composition on a single finite structure.

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$\Sigma_1\mathrm{CoR}$: organized as a fooling-set argument

Before this work, exponential lower bounds were known only for $\Sigma_1\mathrm{CoR}$, even with shared subterms (positive CoR Nakamura 2020; $\Sigma_1\mathrm{CoR}$ Nakamura 2022, Theorems 6.1 and 6.12).

$\Sigma_1\mathrm{CoR}$: complement may be applied only to input relations and $\mathbf{id}$, and so never stands above a composition.

Communication complexity

$F \subseteq f^{-1}(1)$ is a **fooling set** if any two $(a, b), (a', b') \in F$ have a crossed input with value 0.
Covering F by 1-rectangles takes $\geq |F|$ of them.

Nakamura's instances

$t = \Theta(2^k)$ structures $\mathcal{S}_i$ on u_i, w_i, v_i; one $\mathcal{N}$ with u^*, v^*:

- $[\![\varphi]\!]_{\mathcal{S}_i}(u_i, v_i) = 1$, $[\![\varphi]\!]_{\mathcal{N}}(u^*, v^*) = 0$;
- for $i < j$, some c with $(\mathcal{S}_i; u_i, w_i) \hookrightarrow (\mathcal{N}; u^*, c)$ and $(\mathcal{S}_j; w_j, v_j) \hookrightarrow (\mathcal{N}; c, v^*)$.

- $\Sigma_1\mathrm{CoR}$ is **preserved under extensions**: $\iota([\![T]\!]_{\mathcal{S}}) \subseteq [\![T]\!]_{\mathcal{S}'}$ for every embedding ι. $(\diamond)$
- A lowest subterm separating $\mathcal{S}_i$ from $\mathcal{N}$ is a composition $X \,;\, Y$ with witness w_i: a 1-rectangle.
- If $\mathcal{S}_i, \mathcal{S}_j$ share it, c is a witness for (u^*, v^*) in $\mathcal{N}$ by $(\diamond)$. $\Rightarrow$ t compositions.

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With complement, both ingredients fail

1. **Complement reverses** ($\diamond$): $\overline{R\,;\mathbf{1}}$ holds at (u, v) when u has no R-successor, but fails at $(\iota(u), \iota(v))$ once $\iota(u)$ gains one.

2. **A term with two compositions separates every $\mathcal{S}_i$ from $\mathcal{N}$:**

$$\overline{(E \cap (E\,;\overline{\mathbf{id}}))\,;\mathbf{1}} \qquad (E\text{: the adjacency relation})$$

Other lower-bound methods do not apply directly either: formula-size and EF/pebble games (G_k needs one quantifier and three variables; its DNF nests no compositions); rank and discrepancy (free Boolean operations defeat measures of a single matrix).

Unlike the inclusion ($\diamond$), an **equality** survives complement.

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Our technique: exact predictions on one reference structure

1. Modify **one finite structure** $\mathcal{M}_0$ in **exponentially many small ways** $\mathcal{M}_\lambda$ $(\lambda \subseteq [k])$, each changing the answer of G_k.

2. Compare a circuit on $\mathcal{M}_\lambda$ with what its values on $\mathcal{M}_0$ **predict**, by an **equality**: only compositions can notice a modification.

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The reference structure

Edge-colored complete graphs $\mathcal{M} = (V, \chi)$, $\chi\colon \binom{V}{2} \to 2^{[k]}$, $\chi(u,u) \coloneqq \mathbf{id}$ (a label, not a color).

on an edge of color μ : $\quad P_i \iff i \in \mu, \qquad Q_i \iff i \notin \mu \qquad (noinputonloops)$

☛ a read edge of color μ accepts exactly the witness edges of color μ^c.

Two-witness property: for distinct u, v and all μ, ν, at least two $z \notin \{u, v\}$ have $\chi(u,z) = \mu$, $\chi(z,v) = \nu$.

Exists for $|V| \geq n_0(2^k)$: a random coloring (Jipsen et al. 1995).

Fix one: the **reference structure** $\mathcal{M}_0$, and a vertex $\hbar$, the **hub**.

(M1) every vertex has edges of every color;

(M2) the value at (u, v) of every relation that a circuit computes on $\mathcal{M}_0$ depends only on $\chi(u, v)$;

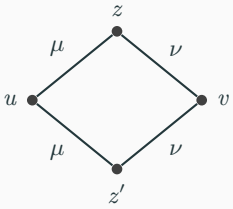
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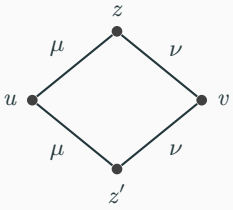
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Exists for $|V| \geq n_0(2^k)$: a random coloring (Jipsen et al. 1995).

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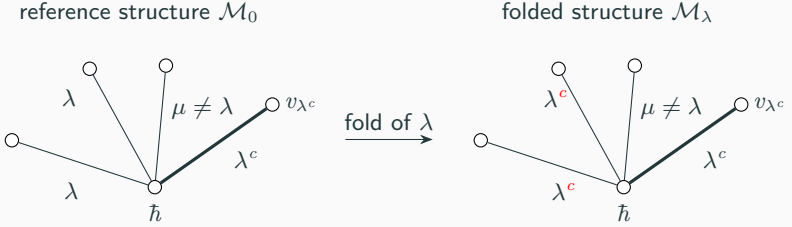
(M1) every vertex has edges of every color;

(M2) the value at (u, v) of every relation that a circuit computes on $\mathcal{M}_0$ depends only on $\chi(u, v)$;

(M3) for distinct $u, v \neq \hbar$ and all μ, ν, some witness $z \notin \{u, v, \hbar\}$ exists.

Folds

The **fold of** λ recolors the edges of color λ at $\hbar$ to λ^c; the result is the **folded structure** $\mathcal{M}_\lambda = (V, \chi_\lambda)$.

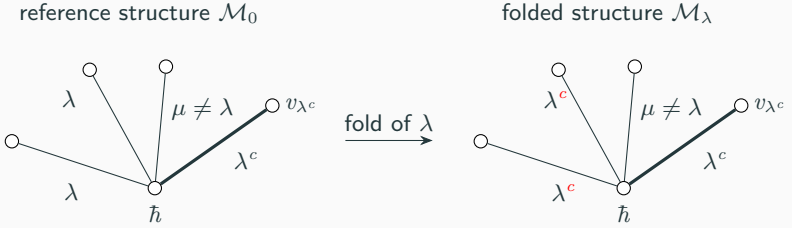


only edges at the hub $\hbar$ are drawn (Figure 1 of the paper)

By (M1) fix v_{λ^c} with $\chi(\hbar, v_{\lambda^c}) = \lambda^c$; the fold does not recolor this read edge. It accepts exactly witness edges of color λ, which $\hbar$ has in $\mathcal{M}_0$ but not in $\mathcal{M}_\lambda$.

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Predictions and noticing

Compare a circuit C on $\mathcal{M}_0$ and $\mathcal{M}_\lambda$ gate by gate:

$$[\![g]\!]_{\mathcal{M}_\lambda}(u,v) = [\![g]\!]_{\mathcal{M}_0}(u',v') \quad \text{whenever } \chi_\lambda(u,v) = \chi(u',v'). \tag{$\star$}$$

By (M2) the right side depends only on the color: what $\mathcal{M}_0$ **predicts** for the new colors.

- every leaf satisfies $(\star)$; an output computing G_k violates it at $(\hbar, v_{\lambda^c})$.

A gate g **notices** the fold of λ if g, applied on $\mathcal{M}_\lambda$ to the values predicted for its operands, does not compute the value predicted for g. This depends only on the operation of g and the values of its operands on $\mathcal{M}_0$.

$$\Delta_g := \{\lambda : g \text{ violates } (\star)\}, \quad D_g := \{\lambda : g \text{ notices the fold of } \lambda\}, \quad \Delta_g \subseteq \Delta_{g_1} \cup \Delta_{g_2} \cup D_g.$$

$$\emptyset = \Delta_{\text{leaf}} \rightsquigarrow \cdots \rightsquigarrow \Delta_{\text{output}} = 2^{[k]} \quad \Rightarrow \quad \text{every color lies in some } D_g.$$

- Unlike $(\diamond)$, the equality $(\star)$ **survives complement**: Boolean operations and converse notice nothing.
- Each composition notices at most two folds (next) $\Rightarrow$ $g_\cdot(C) \geq 2^k/2$; D_g is attached to the gate, so this holds **with sharing**.

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Values on the reference structure are color tables

X has a **color table**: $X(u, v)$ depends only on $\chi(u, v)$; $X[\mu]$, $X[\mathbf{id}]$; $S_X := \{\mu : X[\mu] = 1\}$.

X	$X[\mathbf{id}]$	$X[\{1\}]$	$X[\emptyset]$
P_1	0	1	0
Q_1	0	0	1
$P_1 \,;\, Q_1$	0	1	1
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$k = 1$

Boolean operations and converse act per color. For composition, by two witnesses:

$$(X \,;\, Y)[\mu] = (X[\mathbf{id}] \wedge Y[\mu]) \vee (X[\mu] \wedge Y[\mathbf{id}]) \vee (S_X \neq \emptyset \wedge S_Y \neq \emptyset),$$
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$[\![H_\lambda \,;\, \mathbf{1}]\!]_{\mathcal{M}_\lambda}$ fails at every $(\hbar, v)$, but holds at the other pairs of the same colors. $\Rightarrow$ compare each $\mathcal{M}_\lambda$ with $\mathcal{M}_0$ **directly**.

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Each composition notices at most two folds

Lem. (Each composition notices at most two folds)

For any two relations X, Y on the reference structure that have color tables, $(X\,;\,Y)^{(\lambda)} = X^{(\lambda)}\,;\,Y^{(\lambda)}$ for all colors λ except at most two.

A single pair of color tables; no circuit is involved.

On an arbitrary structure

$R = \{(a, b_i)\}_{i \le t}$, $S = \{(b_i, c_i)\}_{i \le t}$: removing (a, b_i) from R removes (a, c_i) from $R\,;\,S$.
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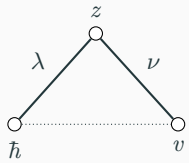
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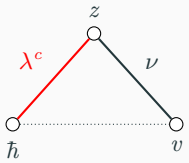
At the hub, and away from the hub

(a) $\mathcal{M}_0$



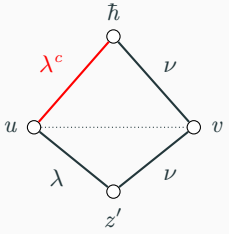
$(X\,;\,Y)(\hbar, v) = 1$

(b) $\mathcal{M}_\lambda$



$(X\,;\,Y)^{(\lambda)}(\hbar, v) = 1,$
$(X^{(\lambda)}\,;\,Y^{(\lambda)})(\hbar, v) = 0$

(c) $\mathcal{M}_\lambda$, away from $\hbar$



$(X\,;\,Y)^{(\lambda)}(u, v) = 1,$
$(X^{(\lambda)}\,;\,Y^{(\lambda)})(u, v) = 1$

$X \coloneqq H_\lambda$, $Y \coloneqq H_\nu$ $(\nu \neq \lambda)$: the composition notices the fold of λ at $(\hbar, v)$, but not away from the hub (Figure 2 of the paper)

Proof of Lemma 1.4

Endpoint witnesses contribute the same to both sides; only outside witnesses matter. The two sides can differ only

pair	why	condition
$(\hbar, v)$	edges $\{\hbar, z\}$ carry every color except λ	$S_X = \{\lambda\}$
$(u, \hbar)$	symmetric	$S_Y = \{\lambda\}$
$(\hbar, \hbar)$	a witness uses the single edge $\{\hbar, z\}$ twice	$S_X \cap S_Y = \{\lambda\}$
$u, v \neq \hbar$	(M3): a witness $z' \notin \{u, v, \hbar\}$	never

Each condition determines λ; if $S_X \cap S_Y \neq \emptyset$, all force $S_X \cap S_Y = \{\lambda\}$. $\Rightarrow$ at most **two** folds. (At most one for a row projection $X\,;\mathbf{1}$.)

Outline

1. The question
2. Main results
3. Natural approaches that fail
4. Technical overview
5. Further results
6. Conclusion

One argument, many targets

Only two ingredients change: how many folds one operation can notice, and the tests of the query.

target or setting	folds noticed per gate	lower bound for G_k
relational circuits	composition 2, projection 1	$g_;(C) \geq 2^{k-1}$
without converse	composition 1 (directed input)	$g_;(C) \geq 2^k$, exact
CoR equations for $G_k^\forall$	as for circuits	$2^{k-1}-1$ compositions
FO^2 formulas, sentences	quantifier 1	2^k quantifiers
matrix languages, breadth β	matrix–matrix 3β, matrix–vector β	$2^k/(3\beta)$ such products
with closure X^*	closure 1	$2g_;(C) + g_*(C) \geq 2^k$
randomized, error ε	as for circuits	factor $1-2\varepsilon$
random inputs	as for circuits	success $\leq \frac{1}{2} + g/2^k + o(1)$, exact

- **Three versus two variables**: $G_k^\forall \coloneqq \forall x \forall y\,(x = y \vee G_k(x,y))$ is FO^2-definable, but needs 2^k quantifiers in FO^2.
- **Arity elimination in matrix query languages**, asked by (Brijder et al. 2019): exponential over every commutative idempotent semiring of finite join breadth.

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Summary

Question	CoR evaluates FO^3 by matrix multiplication; is Tarski's $2^{O(\lvert\varphi\rvert)}$ translation necessary?
Answer	Yes: G_k (size $8k$) needs $\geq 2^{k-1}$ compositions, even for circuits with complement and converse.
Natural approaches	fail: complement reverses the inclusion ($\diamond$) of the fooling-set argument for $\Sigma_1\mathrm{CoR}$.
Our technique	Exact predictions on one reference structure: 2^k folds at the hub; an equality survives complement; each composition notices ≤ 2 folds.

Open questions

- A superpolynomial gap over a **fixed** signature, even one binary relation symbol?
- Domain size: close $\Omega(\sqrt{n/\log n}) \leq c_n(G_k) \leq 4kn$ for $n < n_0(2^k)$.
- The exact composition count for G_k: between 2^{k-1} and 2^k.

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